Technical Data sheet

SLS LoRa Concentrator Board

Author: Kamesh Lakhwani

Date: 07/07/2021

Version / Status: 1.0

Document Library Approver: Lalit Shah

Document Classification: Internal

Distribution and approvals are maintained in:
It is the responsibility of the user of this document to verify that it is the most current edition.

Preface

Privacy information

This document may contain information of a sensitive nature. This information should not be given to any persons outside of the System Level Solutions (I) Pvt. Ltd. without prior consent.

Revision history

Name	Description of Change	Date	Version
Kamesh Lakhwani	Initial version	07/07/2021	1.0

Approval

Name	Position	Signature	Date
Lalit Shah	Manager	Email approval received (must be stored in document library with Approved version)	08/07/2021

Table of Contents

1. Introduction

The Concentrator module is targeted for a huge variety of applications like Smart Metering, IoT and M2M applications. It is a multi-channel high performance transmitter/receiver module designed to receive several LoRa packets simultaneously using different spreading factors on multiple channels. The concentrator module can be integrated into a gateway as a complete RF front end of this gateway. It provides the possibility to enable robust communication between a LoRa gateway and a huge amount of LoRa end-nodes spread over a wide range of distance. The concentrator needs a host system for proper operation. This host system can be a PC or MCU that will be connected to the concentrator via SPI-Interface.

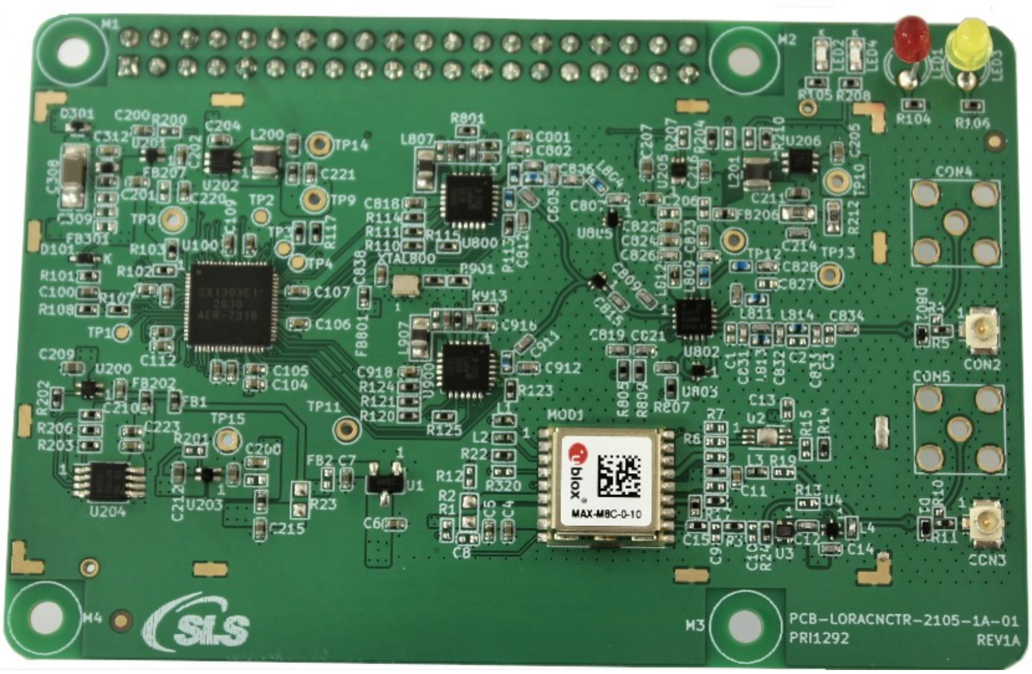

Figure 1-1: Image of the Concentrator Board

The Concentrator is able to receive up to 8 LoRa® packets simultaneously sent with different spreading factors and also on different channels. This unique capability allows to implement innovative network architectures advantageous over other short range systems:

- End-point nodes (e.g; sensor nodes) can change frequency with each transmission in a random pattern. This provides vast improvement of the system robustness in terms of interferer immunity and radio channel diversity.

Version: 0.1
Document Classification: Internal

- End-point nodes can dynamically perform link rate adaptation based (by adapting their spreading factors) on their link margin without adding complexity to the protocol. There is no need to maintain a table of which end point uses which data rate, because all data is demodulated in parallel.
- The capacity of the air interface can be increased due to orthogonal spreading factors.
- Due to the high range a star topology can be used. This results in simple implementation avoiding complex network layers, wireless routers and additional network protocol traffic.

1.1. Key Features

- Compact size 79.8 x 67.3 mm
- LoRa® modulation technology
- Frequency band 868 MHz and 915 MHz Support
- Sensitivity down to -137 dBm
- SPI interface
- SX1303 base band processor
- High-speed 250 / 500 kHz LoRa demodulator
- Multi-SF 125 kHz LoRa® reception with Fine Timestamp
- 1 (G)FSK demodulator
- 2 x SX1250 Tx/Rx front-ends
- Supply voltage +3.3V and/or +5V
- RF interface optimized to 50 Ω
- Output power level up to 20 dBm
- GPS receiver
- Range up to 15 km (Line of Sight)
- Range of several km in urban environment
- Status LEDs
- Standard Raspberry Pi compatible 20 x 2 Connector.

- HAL is available from https://github.com/Loranet/lora_gateway

1.2. Applications

- Smart Metering
- Wireless Star Networks
- Home-, Building-, Industrial automation
- Remote Control
- Wireless Sensors
- M2M, IoT
- Wireless Alarm and Security Systems
- LoRaWAN, etc.

2. LoRa Modulation Technique

The Concentrator uses Semtech's LoRa® spread spectrum modulation technique. This modulation, in contrast to conventional modulation techniques, permits an increase in link budget and increased immunity to in-band interference.

LoRa also provides significant advantages in both blocking and selectivity, solving the traditional design compromise between range, interference immunity and energy consumption, please refer to [1].

Semtech's LoRa® technology transceivers support several bandwidth options and spreading factors ranging from 7 to 12. The spread spectrum LoRa® modulation is performed by representing each bit of payload information by multiple chips of information. The rate at which the payload information is sent is referred to as the nominal symbol rate (Rs), the ratio between the nominal symbol rate and chip rate is the spreading factor and represents the number of modulation symbols sent per bit of information. Note that the spreading factor must be normally known in advance on both transmit and receive sides of the radio link as different spreading factors are orthogonal to each other. Note also the resulting signal to noise ratio (SNR) required at the receiver input. It is the capability to receive signals with negative SNR that increases the sensitivity, so link budget and range, of the LoRa receiver.

For further information on LoRa® please refer to [2].

3. Module Overview

An overview about designation of the key components is given by the following picture

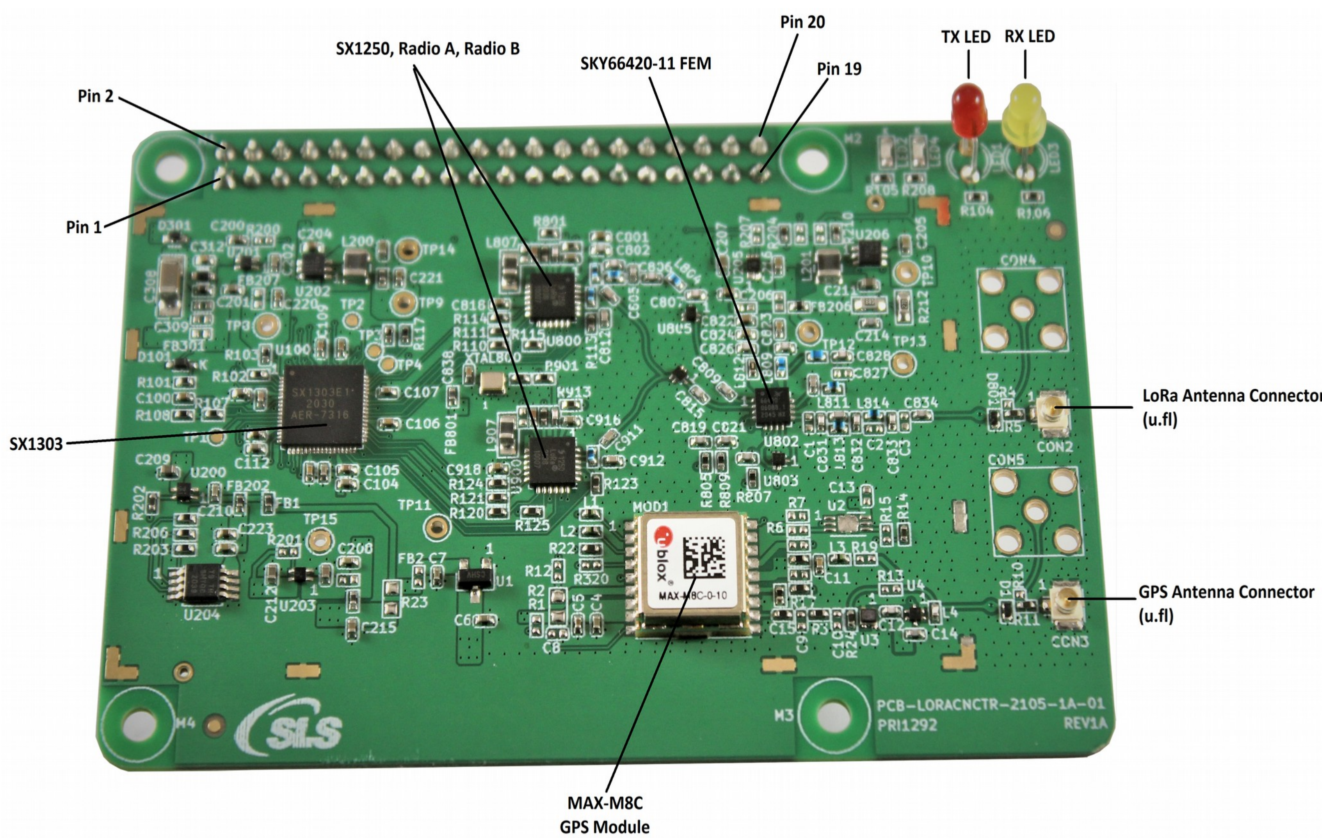

Figure 3-1: Component Overview of the Concentrator

3.1. SX1303

The SX1303 is a new generation of baseband LoRa® chip for gateways. It excels in reducing current consumption, simplifies the thermal design of gateways, and reduces the Bill Of Materials costs, yet it is capable of handling a higher amount of traffic than preceding devices.

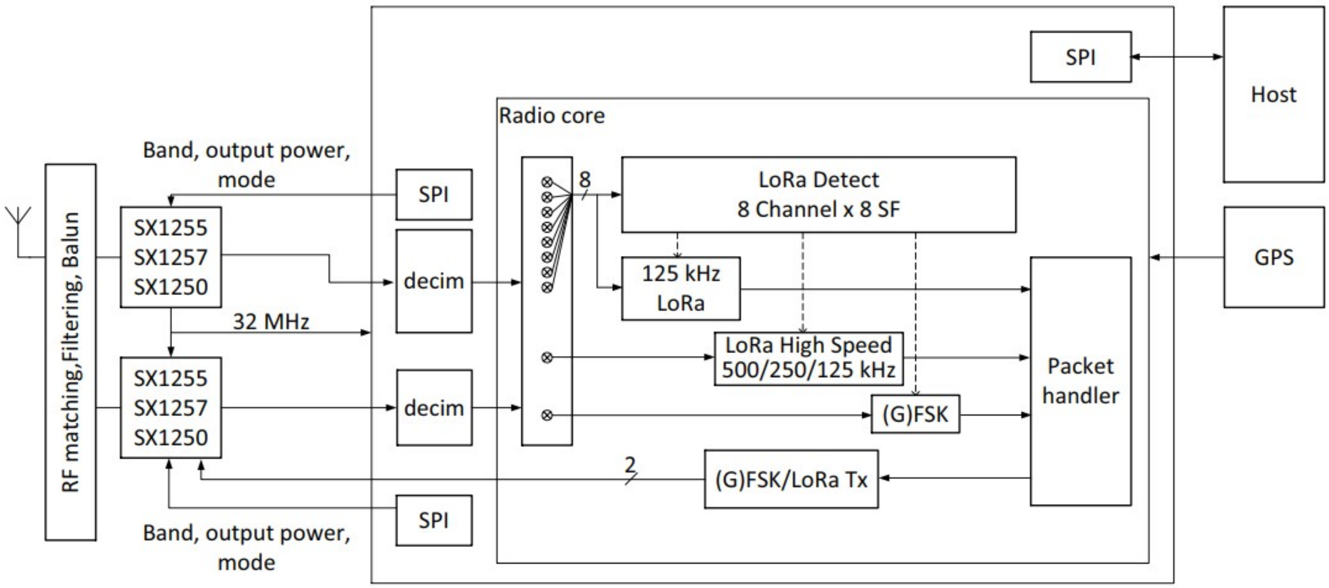

Figure 3-1-1: Block Diagram of the Concentrator with SX1303 Baseband Processor

The high-speed baseband digital engines are clocked from a single 32 MHz clock source, and the chip embeds the capability to support SF5 and SF6 unlike previous generations. The architecture has been reworked to reduce power consumption very significantly; it makes it easier to embed the SX1303 in highly-integrated environments where power dissipation might be a challenge. The SX1303 supports Fine Timestamp capability, when network-based geo-location is required.

The SX1303 is a digital baseband engine, capable of detecting and demodulating large amounts of LoRa® packets expected in the IOT networks. It is intended to be used with various RF Front End chips (RF to IQ), such as Semtech’s SX1255/57/50. The SX1303 is supplied on two different domains, 1.2V for the core of the baseband processing, and 3 to 3.6V for the host and RF interface. The SX1303 can detect at any time, any packet in a combination of 8 different spreading factors (SF5 to SF12) and 10 channels. The control of the SX1301 by the host system (PC, MCU) is made using a Hardware Abstraction Layer (HAL).

Version: 0.1
Document Classification: Internal

3.2 RF Front End Interface

The SX1303 accommodates SX1250 RF front-end device which is one from Semtech products.

The role of these devices is to down-convert the RF signal to baseband (direct conversion or low-IF), and digitize it to feed the IQ samples to the SX1303 baseband chip.

The interconnection to the front-end device is organized as follows.

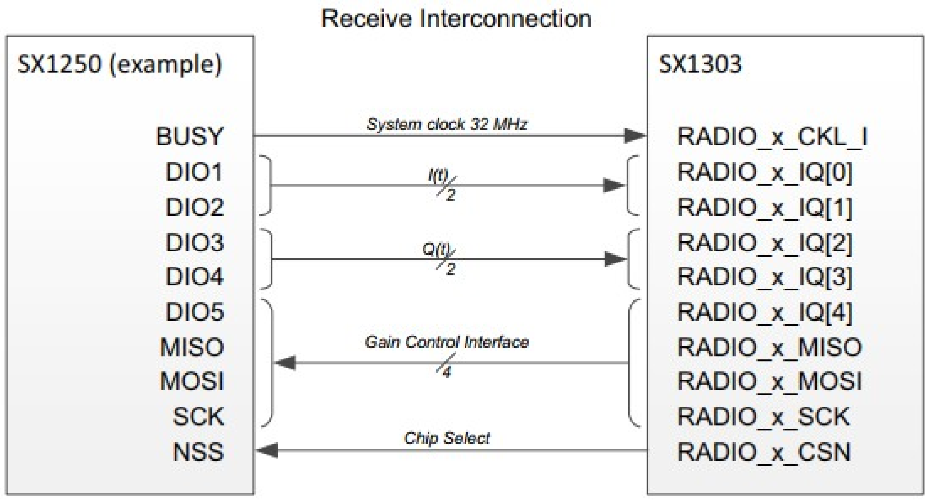

Figure 3-2-1: Receive Interconnection

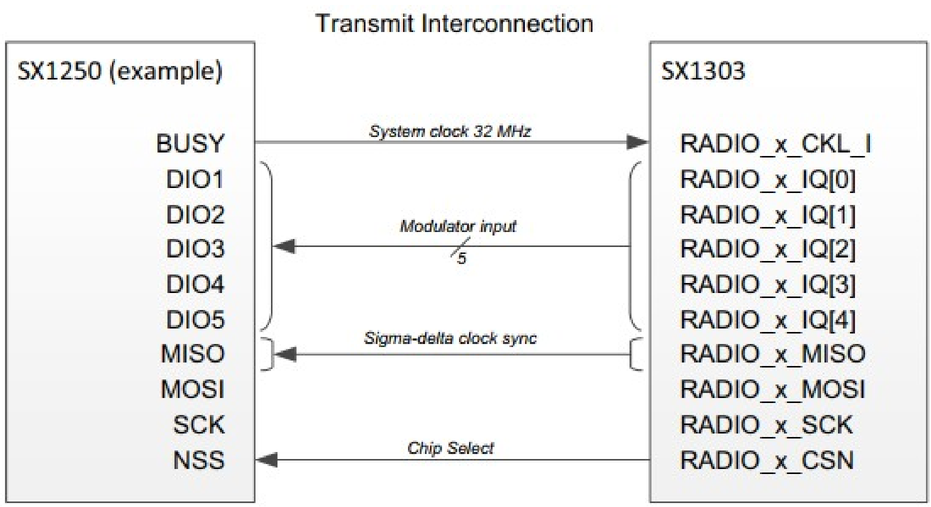

Figure 3-2-2: Transmit Interconnection

3.3 Detection Engine – Modems

The SX1303 offers Fine Timestamp capability, which can be used with Semtech's differential time of arrival technology. A holistic detection engine can capture any LoRa® traffic in the pre-defined frequency plan, assigning the detected packets to the pool of available modems for demodulation. Two modes are proposed for the modem assignment strategy, when timestamps are used:

- Higher Capacity:

Up to 8 packets at any given time can be received for any Spreading Factor. These packets are timestamped, with the exception of packets at SF11 and SF12 (data only).

- Timestamp for all Spreading Factors:

Up to 8 packets at Spreading Factor SF5-12 can be received at any time, including, at most, 4 packets at SF11 and/or SF12. All these packets will be timestamped.

1 high-speed multi-BW LoRa® modem (125, 250 or 500 kHz), handling a single declared SF, and 1 FSK modem is also available.

4. Application Information

The design represents a compact reference implementation of the SX1303, along with its power management, clocks, Front End Module, RF matching and filtering.

4.1. Geographical Designs

The suitability of the SX1303-based reference designs to national radio frequency regulations depends on the RF front-end device being used. With the SX1250/55/57 front-ends provided by Semtech, the expectation is:

- Up to +27 dBm supported in the USA, Canada or other FCC-type countries
- Up the +27 dBm in Europe and other ITU 1 regions
- Up to +21 dBm in Japan where the phase noise requirement is more stringent

4.2. Reference Block Diagram

The application block diagram is shown below:

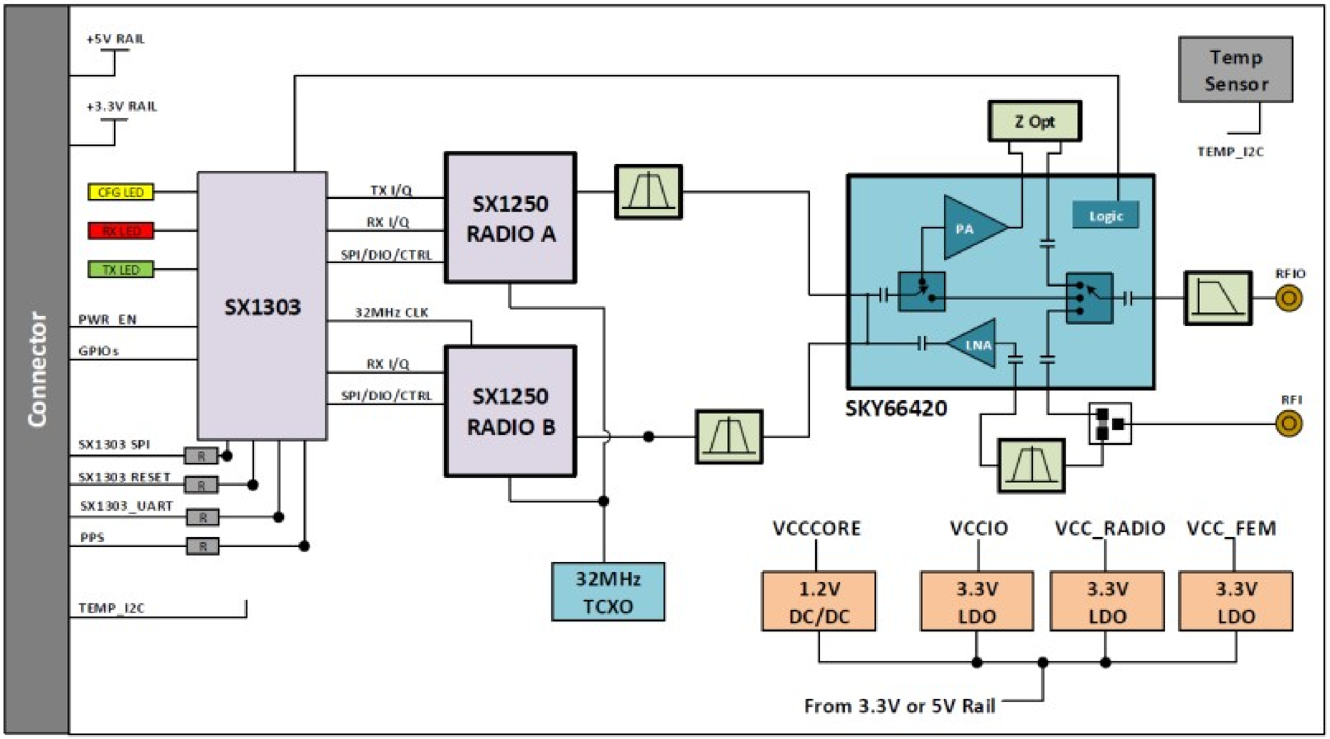

Figure 4-2-1: Application Design Block Diagram

4.3. RF Interface

The Concentrator has two different versions of the board which supports an RF interface for the 868 MHz & 915 MHz frequency band respectively. By connecting an appropriate antenna to the antenna connector, the concentrator is fully ready for communication.

4.4. External Module Connector

For easy integration into a target system and mounting of the concentrator on a carrier board, the headers on the module's bottom side can be used for these purposes (refer to the Table for the pin description).

4.4.1. SPI

The connector on the bottom side provides an SPI connection, which allows direct access to the Sx1301 SPI interface. This gives the target system the possibility to use existing SPI interfaces to communicate to the concentrator board.

After powering up the Concentrator it is required to reset SX1303 via PIN 13, refer to Table.

4.4.2. GPS PPS

In case of available PPS signals in the target system, it is possible to connect this available signal to the appropriate pin at the connector.

4.4.3. UART

The bottom connector provides a UART interface. This interface is for future use.

4.4.4. Digital IOs

There are two GPIOs of the Sx1301 available, which gives the user some possibilities to get information about the system status.

4.4.5. LEDs

Various LED Indications are available on board for below functions:

- TX Packet
- RX Packet
- Power
- SX1303 Configuration

4.4.6. Others

- 1 x I2C : coming from host to the temperature sensor I2C interface

- Power Enable line

- SX1302 reset line

5. LoRa Systems, Network Approach

The use of LoRa® technology can be distinguished in “Public” and “Private” networks. In both cases the usage of a concentrator module can be reasonable. Public networks are operator (e.g. telecom) managed networks whereas private networks are individually managed networks.

LoRa networks are typically star or multiple star networks where a gateway relays the packets between the end-nodes and a central network server, see Figure 4-1. For private network approaches the server can also be implemented on the gateway host.

Due to the possible high range the connection between end-nodes and the concentrator is always a direct link. There are no repeaters or routers within a LoRa network.

Depending on the used spreading factor and signal bandwidth different data rates1 (0.3 kbps to ~22 kbps) and sensitivities down to -137 dBm are possible. Spreading factor and signal bandwidth are a trade-off between data rate and communication range.

5.1. Overview

The Concentrator is able to receive on different frequency channels at the same time and is able to demodulate the LoRa signal without knowledge of the used spreading factor of the sending node.

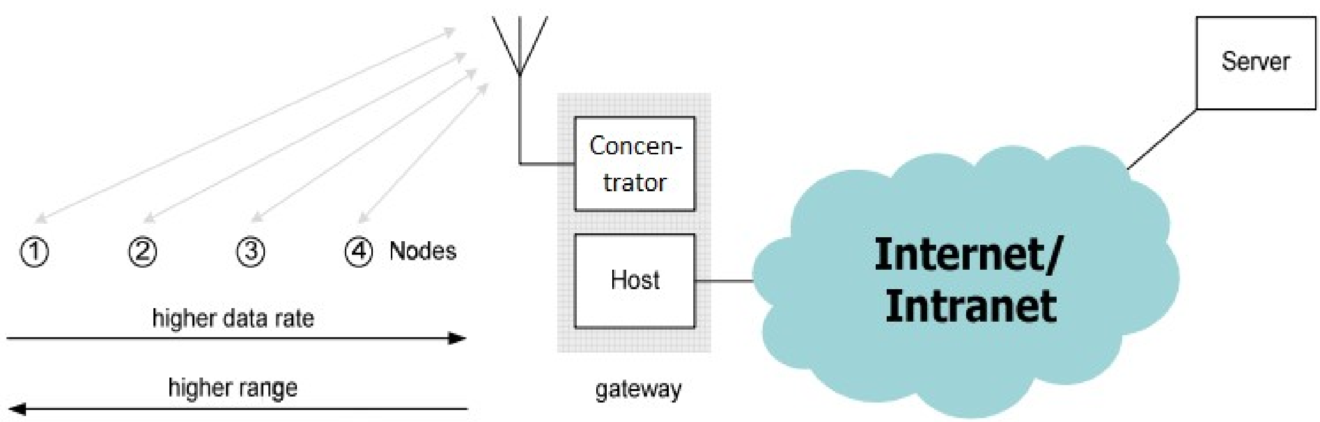

Figure 5-1-1: Public LoRa Network Approach

Due to the fact that the combination of spreading factors and signal bandwidths results in different data rates the use of “Dynamic Data-Rate Adaption” becomes possible. That means that LoRa nodes with high distances from the Concentrator must use higher spreading factors and therefore have a lower data rate. LoRa nodes which are closer to the concentrator can use lower spreading factors and therefore can increase their data rate.

Due to the fact that spreading factors are orthogonal and Concentrator supports up to 10 demodulations paths the channel capacity of a LoRa cell can be increased using Concentrator compared to conventional modulation techniques.

6. Firmware

The LoRaWAN specification is currently driven by the LoRa Alliance™. Currently all available software, firmware and documentation can be found and downloaded from the open source project LoRa-net hosted on https://github.com/Lora-net

This project considers all parts that are needed to run a network based on LoRa technology. It includes the node firmware (several hardware platforms are supported), the gateway host software (HAL driver for SX1303, packet forwarder) and a server implementation.

7. Electrical Characteristics & Timing specifications

In the following different electrical characteristics of the Concentrator are listed. Furthermore details and other parameter ranges are available on request.

Note: Stress exceeding of one or more of the limiting values listed under “Absolute Maximum Ratings” may cause permanent damage to the radio module.

7.1. Absolute Maximum Ratings

Parameter	Condition	Min	Typ.	Max	Unit
Supply Voltage (VDD)		-0.3	3.3	3.6	V
Operating Temperature		-40		+85°C	°C
RF Input Power					dBm
Max Pin on Hot-Switching RF-Switch					dBm
VSWR on any RF connector					

Note: With RF output power level above +15 dBm a minimum distance to a transmitter should be 1 m for avoiding too large input level.

7.2. Global Electrical Characteristics

T = 25°C, VDD = 3.3V (typ.) if nothing else stated

Parameter	Condition	Min	Typ.	Max	Unit
Supply Voltage (VDD)		3.0	3.3	3.6	V
Receiver Current Consumption	medium activity (2 radios, 4 active paths)				mA
	high activity (2radios, 10 active paths)				

Version: 0.1
Document Classification: Internal

T = 25°C, VCC_IO = 3.3 V (typ.) if nothing else stated

Parameter	Condition	Min	Typ.	Max	Unit
Logic low input threshold (VIL)	"0" logic input	-0.3	-	0.3*VCC_IO	V
Logic high input threshold (VIH)	"1" logic input	0.7*VCC_IO	-	VCC_IO + 0.3	V
Logic low ouput level (VOL)	"0" logic output, Imax = 8mA	0	-	0.4	V
Logic high output level (VOH)	"1" logic output, Imax = -8 mA	VCC_IO - 0.6	-	VCC_IO	V

7.3. SPI Interface Characteristics

All timings are given in next table for Max load cap of 10 pF.

Parameter	Condition	Min	Typ.	Max	Unit
Timing constraints in SPI inputs					
tSPI,SCK	SCK period	100			ns
tSPI,SCKH	SCK high duration	40			ns
tSPI,SCKL	SCK low duration	40			ns
tSPI,SCKR	SCK rise time (10% VCC_IO ->90% VCC_IO)	0	2.5		ns
tSPI,SCKF	SCK fall time (90% VCC_IO ->10% VCC_IO)	0	2.5		ns
tSPI,DELAY	SCK lead time	40			ns
tSPI,QUIET	SCK trail time	40			ns
tSPI,CSH	Time between two successive CSN chip select	250			ns

tSPI,CSR	CSN rise time (10% VCC_IO ->90% VCC_IO)	0	2.5		ns
tSPI,CSF	CSN fall time (90% VCC_IO ->10% VCC_IO)	0	2.5		ns
tSPI,SETUP	Data in setup time	5			ns
tSPI,HOLD	Data in hold time	5			ns
SPI output timing specification					
tSPI,DOEN	SPI output enable time	0		10	ns
tSPI,DODIS	SPI output disable time	0		10	ns
tSPI,DOV	SCK out falling edge to MISO delay			15	ns

7.4. RF Characteristics

7.4.1. Transmitter RF Characteristics

The Concentrator has an excellent transmitter performance, which generally give a lot of possible settings for the power amplifier of the Concentrator. It is highly recommended, to use an optimized configuration for the power level configuration.

The Concentrator is specified for a max. RF output power of +27 dBm. Long-term operating of the Concentrator with more than +27 dBm can destroy the internal power amplifier of Concentrator. Especially in case of operating the Concentrator with the github software it need to be ensured, that the maximum RF output power of +27 dBm is not exceeded. Therefore the settings of the global_conf.json might need to be changed accordingly.

T = 25°C, VDD = 3.3 V (typ.), 866.5 MHz if nothing else stated

Parameter	Condition	Min	Typ.	Max	Unit
Frequency Range	SCK period	863	-	870	MHz

Modulation Techniques	FSK / LoRa™	-	+/-3	-	kHz
TX Frequency Variation vs. Temperature	-5°C to +55°C				dB
TX Power Variation vs. Temperature[1]	Max. power level, -5°C to +55°C				dB
	Max. power level, -40°C to +85°C				dB
TX Power Variation vs. Frequency	Max. power level				dB
TX Power Variation (initial)	Max. power level				dB
TX Current Consumption	Gain setting for nom. +14 dBm -5°C to +55°C PA=2; DAC=3; Mix=10; Dig=2				mA
	Gain setting for nom. +27 dBm -5°C to +55°C PA=2; DAC=3; Mix=12; Dig=0				mA

[1]Operational temperature range can be basically extended to -40°C to +85°C, but a larger power level drift vs. temperature need to be expected. In addition it is recommended, to use optimized TX settings which are also individually adapted to the temperature.

7.4.2. Receiver RF Characteristics

It is highly recommended, to use optimized RSSI calibration values. For both, Radio 1 and 2, the RSSI-Offset should be set to -169.

The following table gives typically sensitivity level of the Concentrator:

Signal Bandwidth/[kHz]	Spreading Factor	Sensitivity/[dBm]
125	12	
125	7	
250	12	

250	7	
500	12	
500	7	

7.4.3. Certification and Compliancy Restrictions

8. Module Package

In the following the Concentrator module package is described. This description includes the Concentrator pinout as well as the modules dimensions.

8.1. Pinout Description

The Concentrator provides headers at the bottom side, which have a pitch of 2.54 mm. The description of the pins is given by Table below. An additional overview gives Figure 8-1-1.

Pin	Pin Name	Pin Type	Description
1	VDD	Power	+3.3V Supply Voltage
2	VDD	Power	Optional +5.0V Supply Voltage(To be used if input supply to the board is +5.0V instead of +3.3V)
3	SDA	I/O	I2C Data line for on-board temperature sensor
4	VDD	Power	Optional +5.0V Supply Voltage(To be used if input supply to the board is +5.0V instead of +3.3V)
5	SCL	Input	I2C Clock line for on-board temperature sensor
6	GND	Power	System Ground
7	NC	NC	Reserved
8	GPS_RX	Output	GPS UART RxD
9	GND	Power	System Ground
10	GPS_TX	Input	GPS UART TxD
11	GPIO6	I/O	SX1303 General purpose IO
12	PPS	Input	Optional Time pulse (1PPS) from GPS
13	Reset	Reset	SX1303 Asynchronous Reset Input
14	GND	Power	System Ground
15	NC	NC	Reserved

16	GPIO8	I/O	SX1303 General purpose IO
17	VDD	Power	+3.3V Supply Voltage
18	GPS_Reset	Input	GPS Reset Pin
19	MOSI	Input	SX1303 SPI-MOSI
20	GND	Power	System Ground
21	MISO	Output	SX1303 SPI-MISO
22	Power_EN	Input	Optional
23	SCK	Input	SX1303 SPI-Clock
24	CS	Input	SX1303 SPI-Chip Select
25	GND	Power	System Ground
26	NC	NC	Reserved
27	NC	NC	Reserved
28	NC	NC	Reserved
29	NC	NC	Reserved
30	GND	Power	System Ground
31	NC	NC	Reserved
32	NC	NC	Reserved
33	SX1261_DIO1	I/O	SX1303 General purpose IO
34	GND	Power	System Ground
35	GPS_ANT_DET	Input	External Interrupt Signal of GPS
36	NC	NC	Reserved
37	NC	NC	Reserved
38	NC	NC	Reserved
39	GND	Power	System Ground
40	VCC_BATT	Power	Optional Backup voltage supply for GPS

Note: MISO-signal is always low impedance. Do not share with other MISO-signal by direct connection.

8.2. Module Dimensions

The outer dimensions of the Concentrator are given by 85.00 X 56.00 SQ MM mm ± 0.2 mm. The Concentrator provide four drills for screwing the PCB to another unit each with a drill diameter of 2.7 mm.

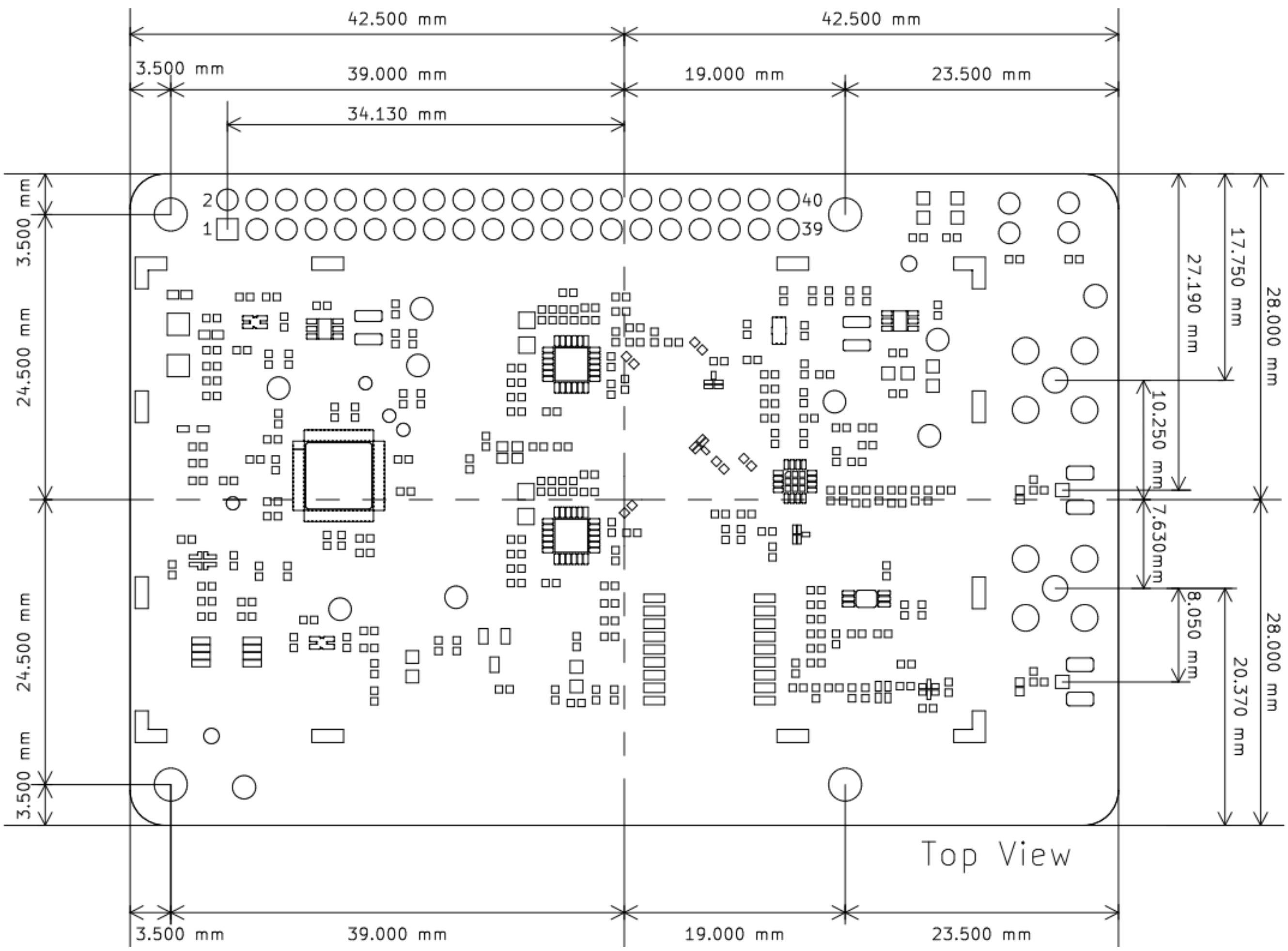

Figure 8-2-1: Concentrator outlines and pins of bottom connector in top view

Version: 0.1
Document Classification: Internal

9. Ordering Information

Ordering Part Number	Description	Distributor
	Concentrator Module with SPI interface	

10. Restrictions and Limitations

10.1. Hardware Restrictions and Limitations

The characteristic values given by the present document are typically obtained by measurements based on evaluation kits of the entitled device. Using other carrier boards or connected equipment might lead to different characteristics. Subject to given measurement results the characteristic values might show the best performance of the entitled device, independent from any compliancy restriction of final operation purposes.

10.2. Software Restrictions and Limitations

The present document is a datasheet of the entitled device which intentional use is to provide information about basic characteristics related to the device hardware. Typically all described characteristic values require software for obtaining them accordingly. All features of the available software are subject to changes without claim to be complete at any time. Characteristically values might also be provided based on datasheets of the appropriate key components unless there are test results available based on the available software.

10.3. Compliancy Restrictions and Limitations

The entitled device has been designed to comply with the standards namely given in the present document. The intentional operation shall be in so called ISM bands, which can be used free of charge within the European Union and typically licences free all over the world. Nevertheless, restrictions such as maximum allowed radiated RF power or duty cycle may apply which might result in a reduction of these parameters accordingly.

In addition, the use of radio frequencies might be limited by national regulations which requirements also need to be met.

In case the entitled device will be embedded into other products (referred as “final products”), the manufacturer for this final product is responsible to declare the conformity to required standards accordingly. A proof of conformity for the entitled device is available from SLS on request. Beside the entitled device the conformity also considers software as well as supporting hardware characteristics which might also have an impact accordingly.

The applicable regulation requirements are subject to change. SLS does not take any responsibility for the correctness and accuracy of the aforementioned information. National laws and regulations, as well as their interpretation can vary with the country. In case of uncertainty, it is recommended to contact either SLS’s accredited Test Center or to consult the local authorities of the relevant countries.

10.4. Disclaimer

SLS points out that all information in this document are given on an “as is” basis. No guarantee, neither explicit nor implicit is given for the correctness at the time of publication. SLS reserves all rights to make corrections, modifications, enhancements, and other changes to its products and services at any time and to discontinue any product or service without prior notice. It is recommended for customers to refer to the latest relevant information before placing orders and to verify that such information is current and complete. All products are sold and delivered subject to “General Terms and Conditions” of SLS, supplied at the time of order acknowledgment. SLS assumes no liability for the use of its products and does not grant any licenses for its patent rights or for any other of its intellectual property rights or third-party rights. It is the customer’s duty to bear responsibility for compliance of systems or units in which products from SLS are integrated with applicable legal regulations. Customers should provide adequate design and operating safeguards to minimize the risks associated with customer products and applications. The products are not approved for use in life supporting systems or other systems whose malfunction could result in personal injury to the user. Customers using the products within such applications do so at their own risk.

Any reproduction of information in datasheets of SLS is permissible only if reproduction is without alteration and is accompanied by all given associated warranties, conditions, limitations, and notices. Any resale of SLS products or services with statements different from or beyond the parameters stated by SLS for that product/solution or service is not allowed and voids all express and any implied warranties. The limitations on liability in favor of SLS shall also affect its employees, executive personnel and bodies in the same way. SLS is not responsible or liable for any such wrong statements.

11. Appendix

11.1. List of Abbreviations

GND	Ground
GPIO	General Purpose Input/Output
GPS	Global Positioning System
HAL	Hardware Abstraction Layer
IF	Intermediate Frequency
IoT	Internet of Things
ISM	Industrial, Scientific and Medical
M2M	Machine to Machine
MCU	Microcontroller Unit
PCB	Printed Circuit Board
PPS	Pulse Per Second
RF	Radio Frequency
SNR	Signal to Noise Ratio
SPI	Serial Peripheral Interface

11.2. References

[1] Semtech, White Paper LoRa Modulation from www.semtech.com

[2] ERC Recommendation 70-03 “Relating to the use of Short Range Devices (SRD)”, Tromsø 1997, CEPT ECC subsequent amendments 13 October 2017

[3] Semtech, SX1303 Data Sheet from www.semtech.com